

Michigan Electrical Grid Demand Fluctuations

Assessing Data Center Feasibility

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Executive Summary

Problem:

- Michigan offering tax incentives for 4-10 new data centers
- Historical grid reliability issues raise sustainability concerns

Approach:

- Linear regression models using 2022-2024 MISO & NOAA data
- Split by temperature (cold <55°F, warm ≥55°F) to capture seasonal patterns

Key Findings:

- Peak demand occurs during high temperature + humidity, June to September
- **Medium data centers (4): Minimal risk**
- **Hyperscale data centers (10): 2.3% added stress at peak (>600k MWh)**
- Model accuracy: $R^2=0.68$ (warm), $R^2=0.35$ (cold)

Recommendation:

- Energy companies should plan increased capacity for summer heat waves with high humidity
- Future studies and further modelling needed!

The Problem

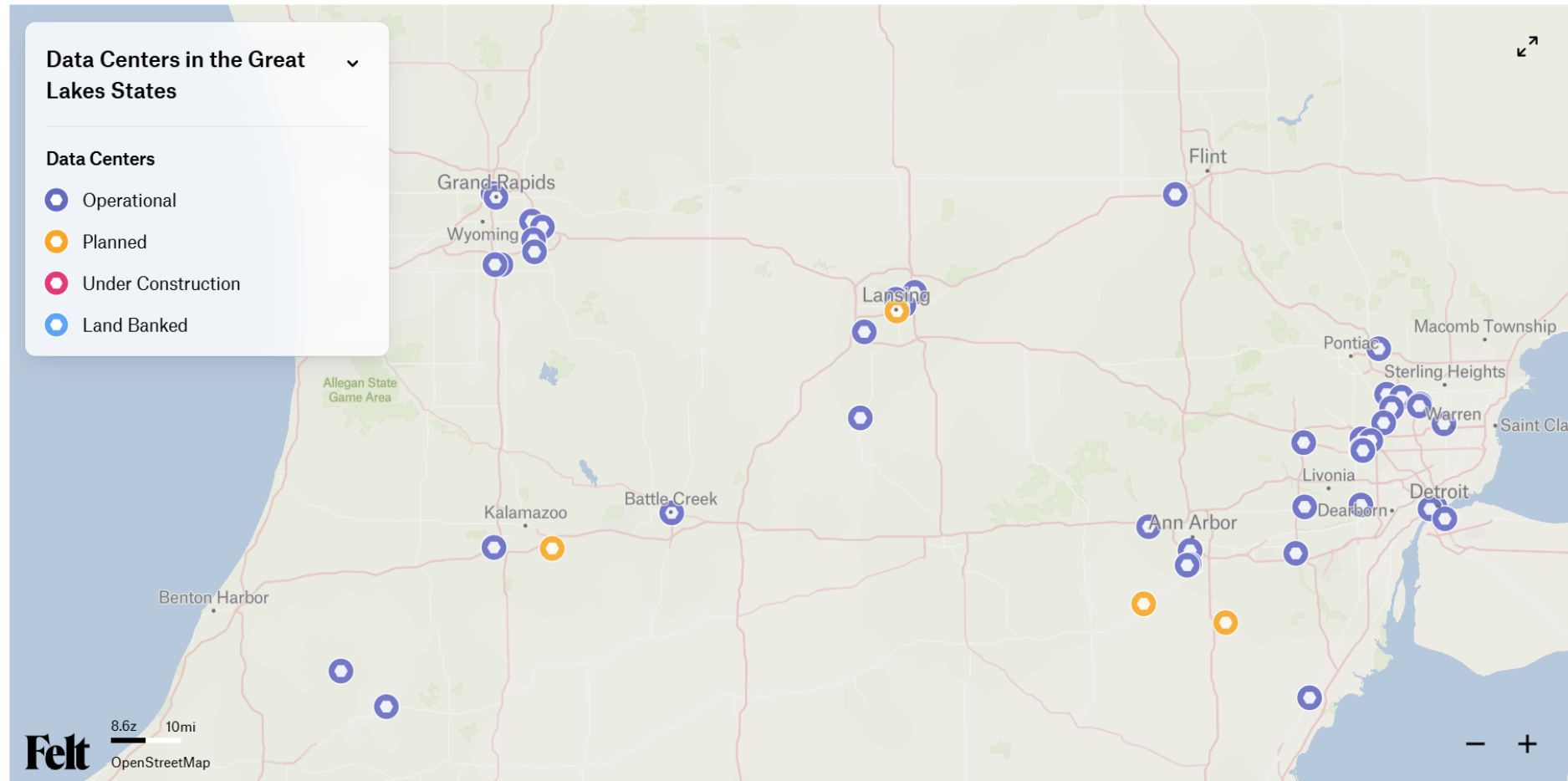
- Michigan is offering tax incentives to attract medium and hyperscale data centers in up to 10 counties [2]
 - Medium data center daily demand: 10-50 megawatt-hours [1]
 - Hyperscale data center daily demand: 100-1,000+ megawatt-hours
- Can Michigan's electrical grid handle both peak seasonal demands and the additional draw caused by new data centers?
 - Under what weather conditions should Consumers Energy and DTE plan additional output?

Proposed Data Centers in Michigan, 2025

Michigan currently has 59 data centers

4 more are planned in Lansing and outside Ann Arbor and Kalamazoo. [1]

6 additional locations are under negotiation [2]



The data for this map provided by <https://www.datacentermap.com/>

Related Work

- **Forecasting Methods:** [4-6]
 - Multiple machine learning algorithms tested for demand forecasting
 - Adaptive Neural-based Fuzzy Inference System (ANFIS) showed highest accuracy using seasonal data and activity patterns
 - Future work: divide aggregated and low-voltage demand; include calendar data
- **Outage Analysis:** [7-10]
 - Metropolitan and well-funded areas show better grid resilience
 - Precipitation and heat most commonly co-occur with 8+ hour outages
 - Future work: Social Vulnerability Index to identify at-risk populations; economic analysis
- **Current Study's Contribution:**
 - Uses linear regression to predict peak demand periods (rather than outages) based on temperature thresholds (rather than calendar seasons)
 - Identifies grid stress points when data center loads added to inform production planning for utilities

Preparing Public Climate Data

- Climate data: NOAA climate data (csv downloaded on Nov. 26, 2025)
 - Representative climate point: the airport weather station in Lansing, MI
- Cleaning:
 - Remove columns without data or outside scope (78 columns removed)
 - Convert date to date type and sort (other data formats acceptable)
 - Fill missing values using linear interpolation and forward fill methods
 - Check for missing measurements (9999), unusual values – none left

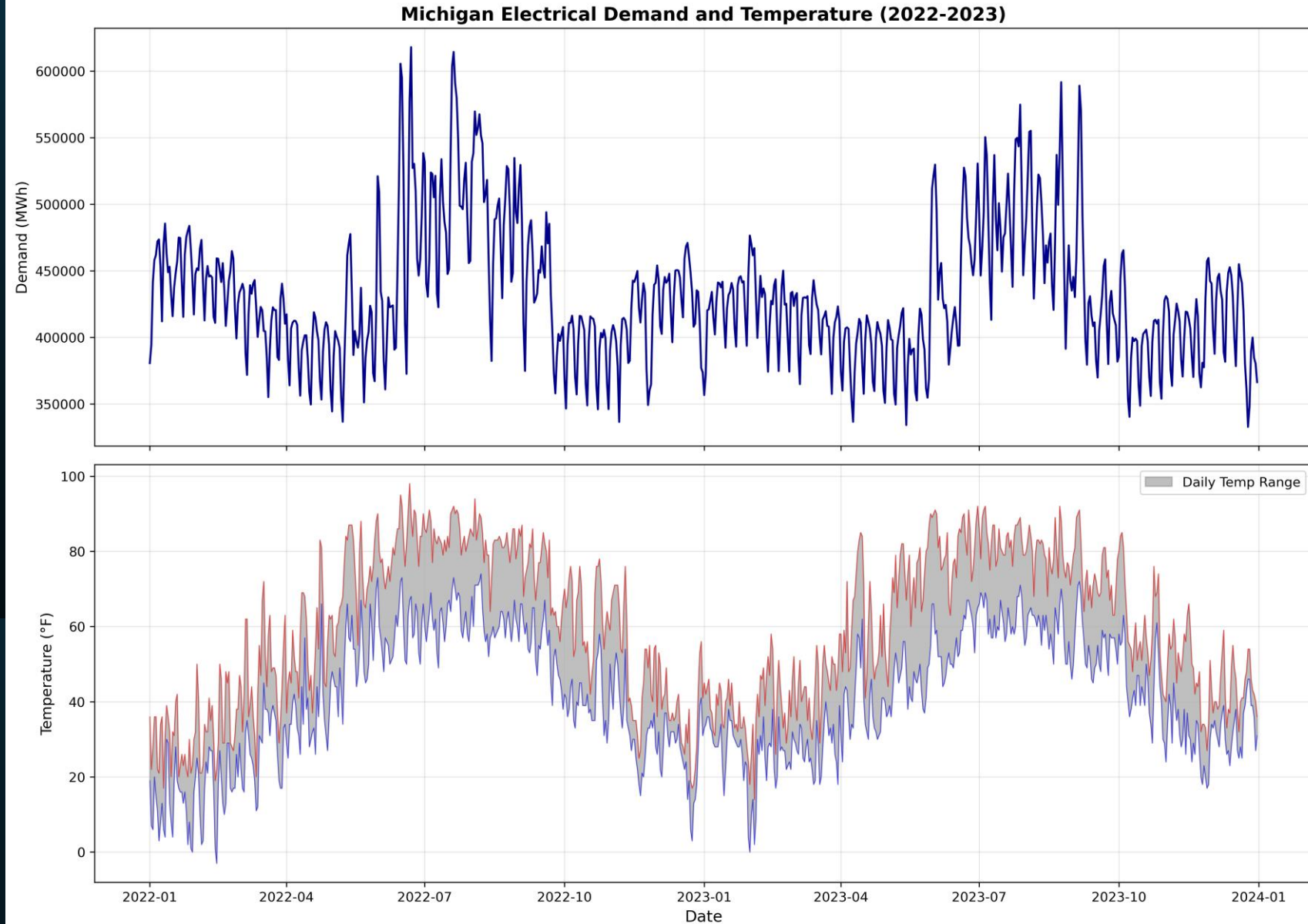
Preparing Electrical Demand Data and Splitting

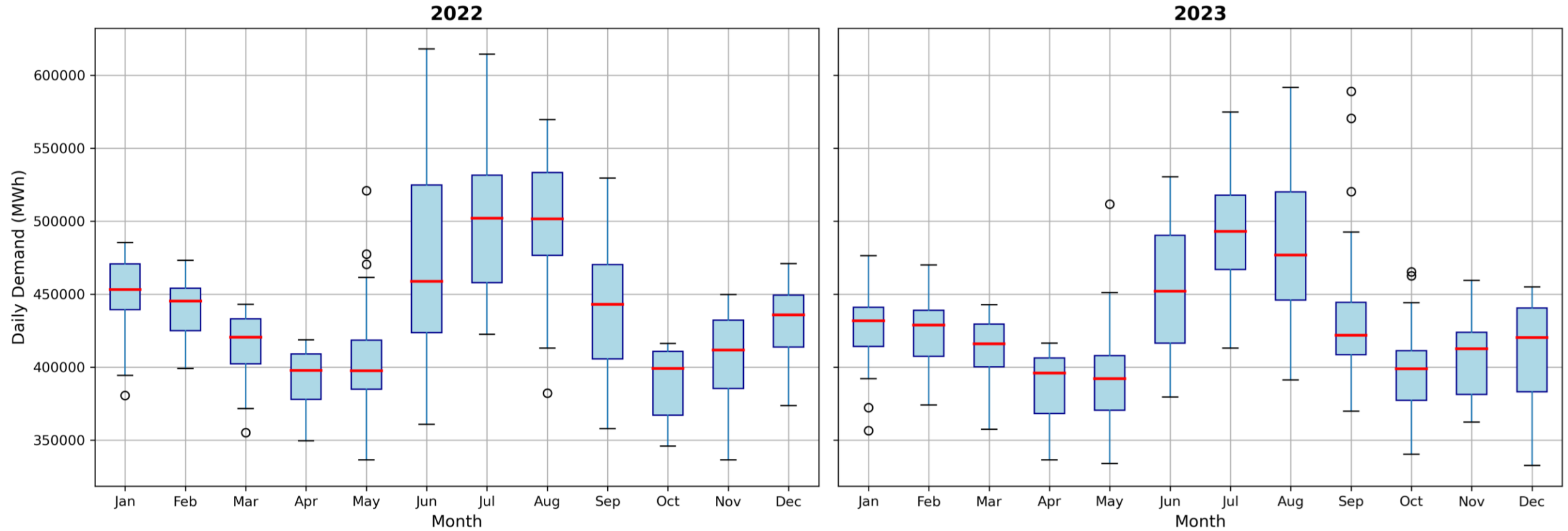
- Electrical data:
 - EIA: US Energy Information Administration (API accessed Nov. 26, 2025)
 - Cleaning:
 - Convert date to date type and sort (other data types acceptable)
 - Remove 7 days without 24 hours of data; no other missing values
 - Aggregate by day to match climate data = 1089 days
- Merge climate and electrical data using inner merge
 - Data shape: 1089 rows (days), 43 columns (attributes and metadata)
- Split for evaluation
 - Training data: 2022-2023
 - Testing data: 2024

2022-2023

Daily Electrical
Demand for the
State of Michigan

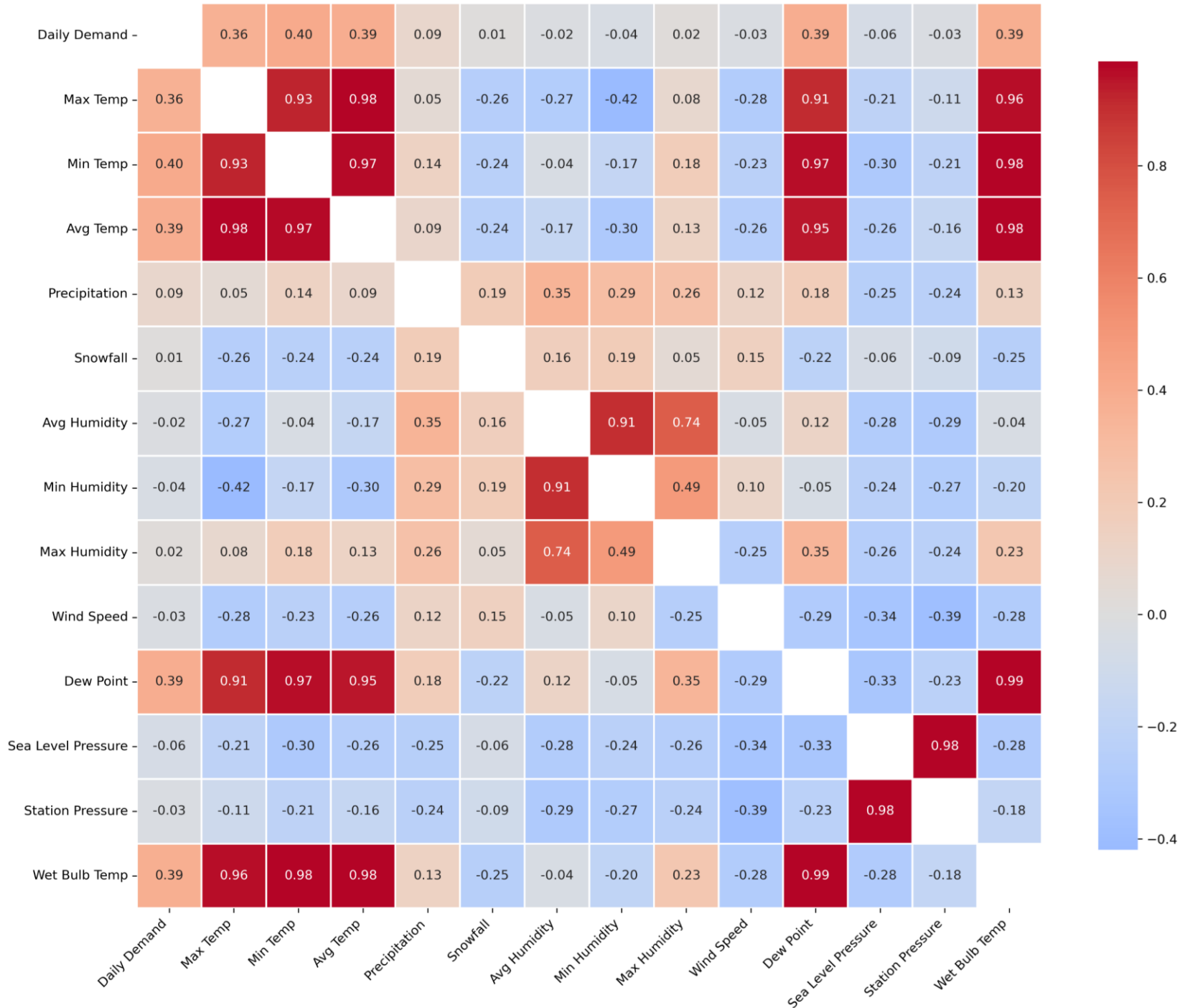
Daily Temperature
Highs and Lows at
the Lansing, MI
Airport



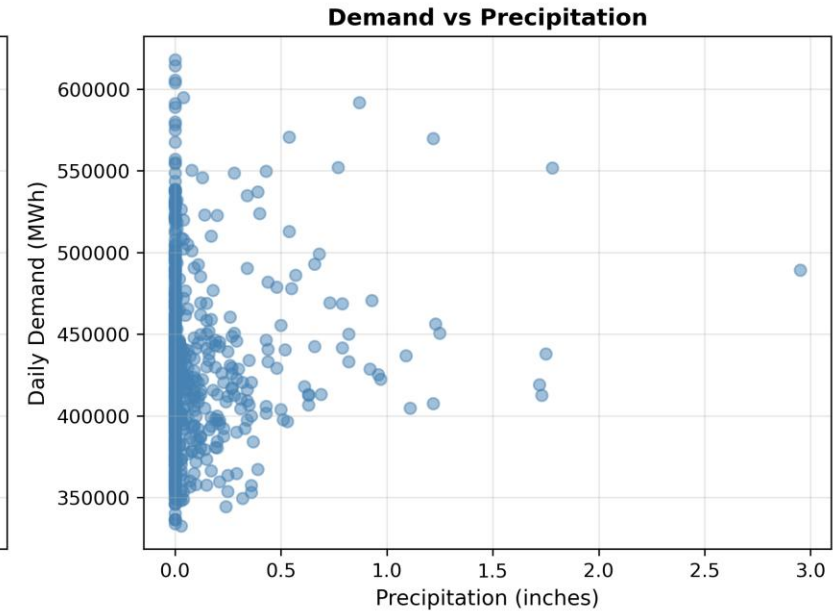
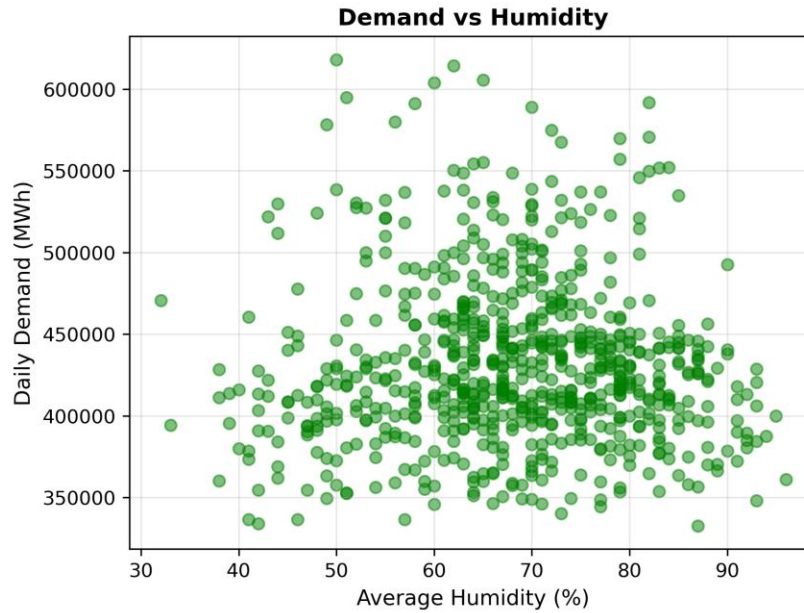
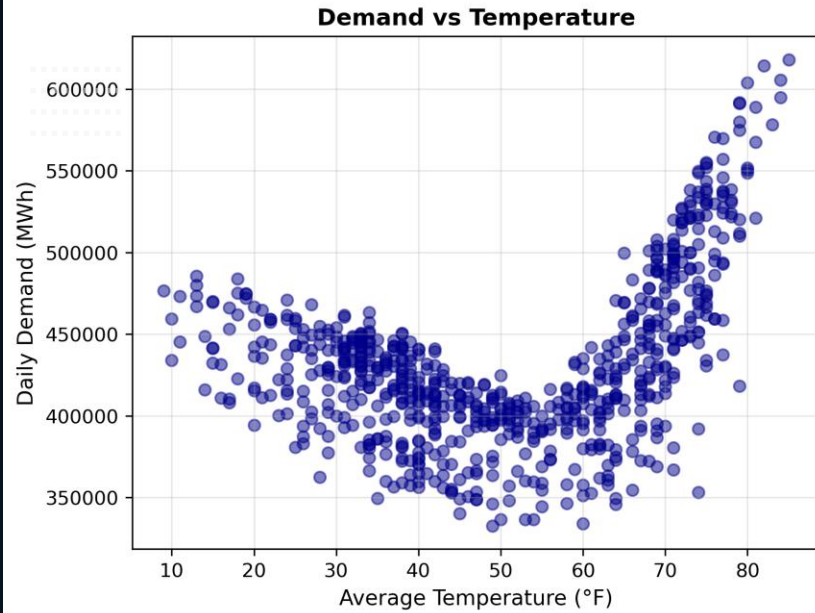


Monthly Electrical Demand for the State of Michigan 2022-2023

Correlation Matrix: Electricity Demand and Weather Variables 2022-23



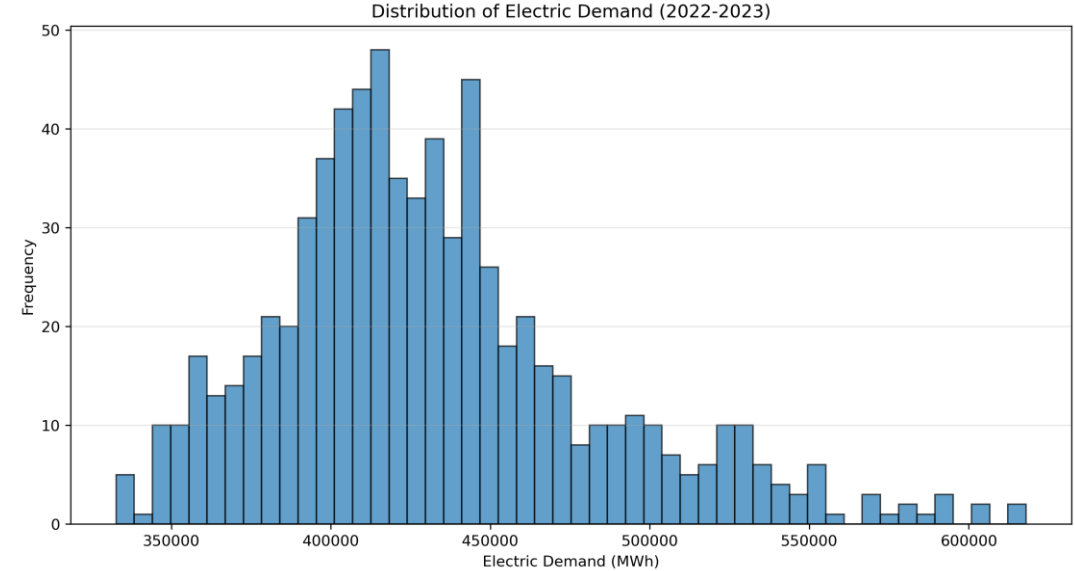
Electrical Demand vs Weather Variables (2022-2023)



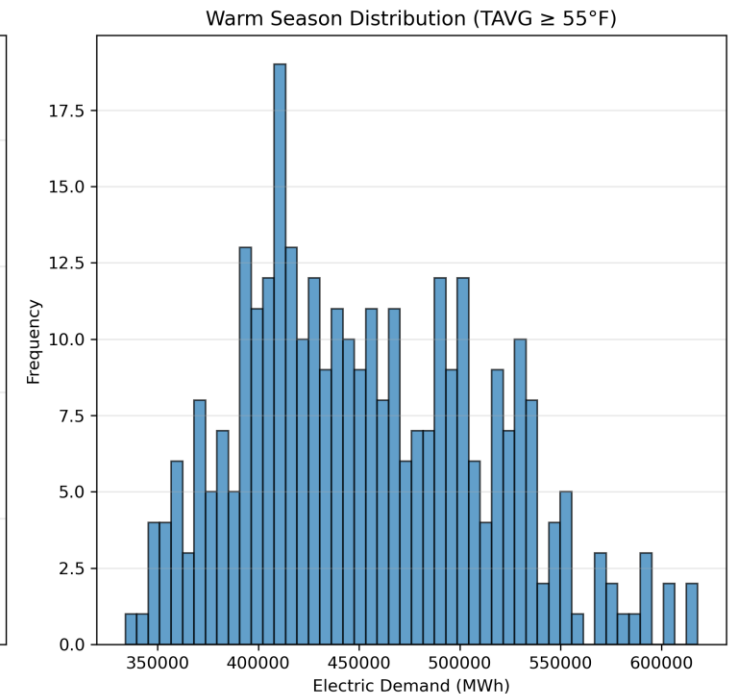
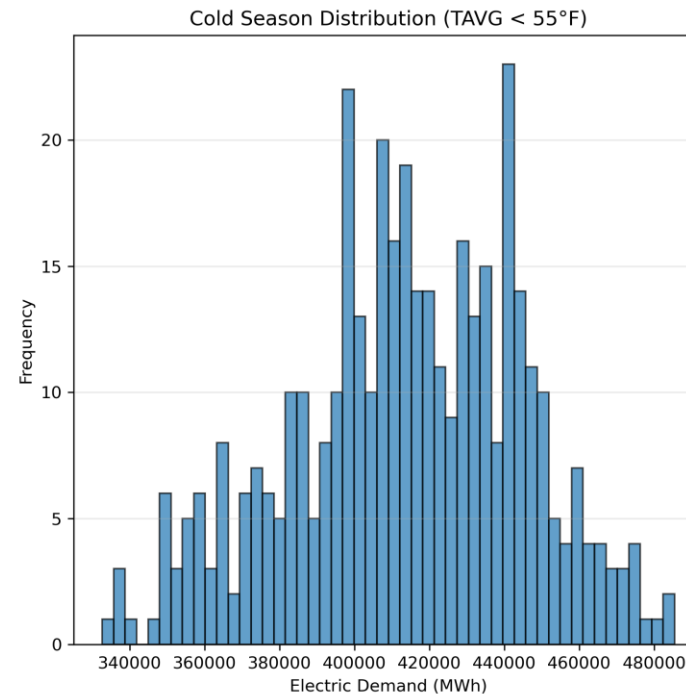
Electrical Demand v. Weather Variables, 2022-2023

- Scatterplots confirm the correlation between demand and temperature → separate linear regressions for high temperatures and low temperatures required
- Humidity and precipitation plots show no discernable correlation

Distribution of Michigan Daily Electrical Demand (2022-2023)



Dividing the data at 55 degrees Fahrenheit creates more normally distributed data sets.



Methods and Python Libraries

- Statsmodels: multiple linear regression model
- Pandas: data manipulation
- NumPy and SciPy: number crunching and stats tests
- Matplotlib and seaborn: visualizations

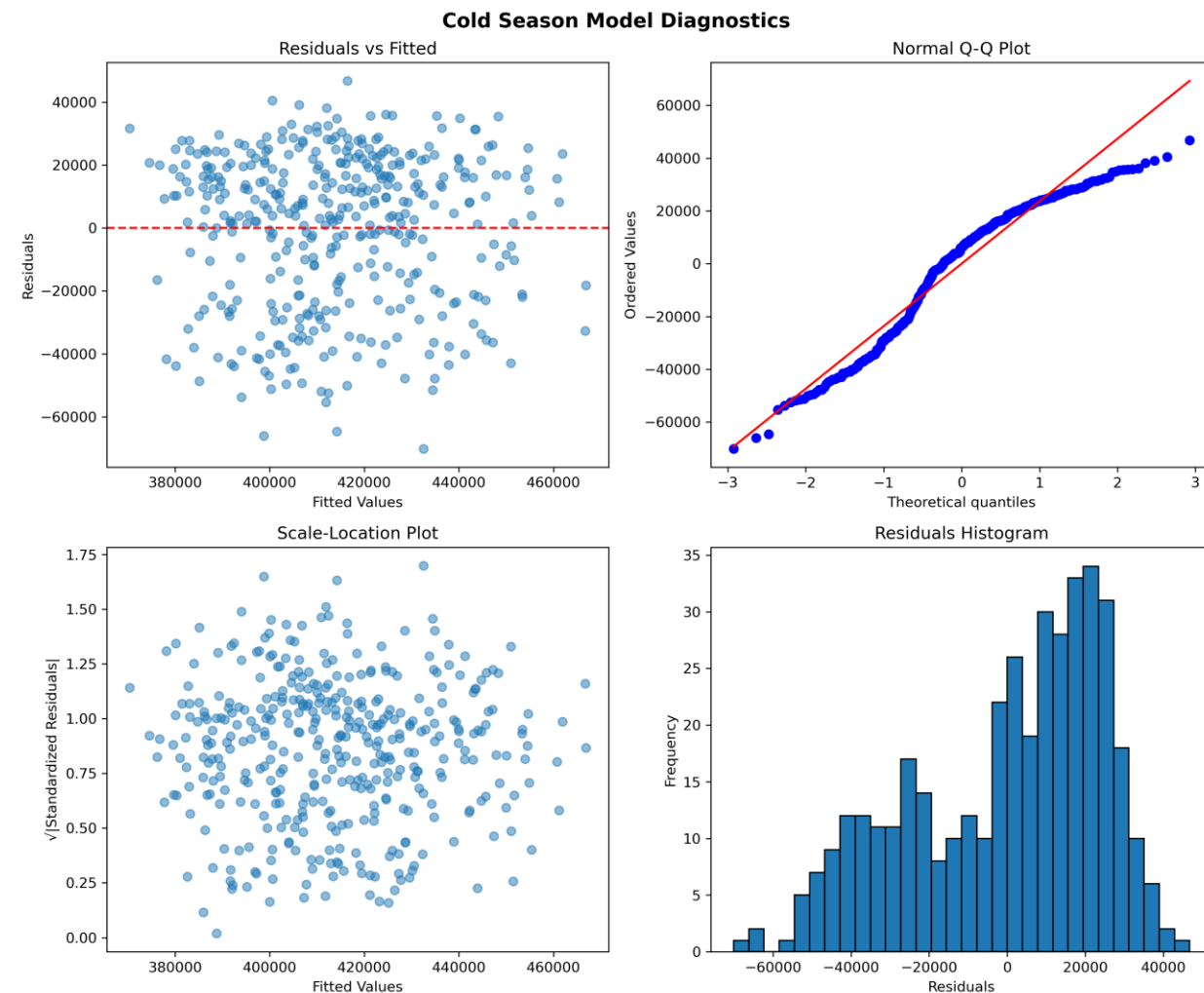
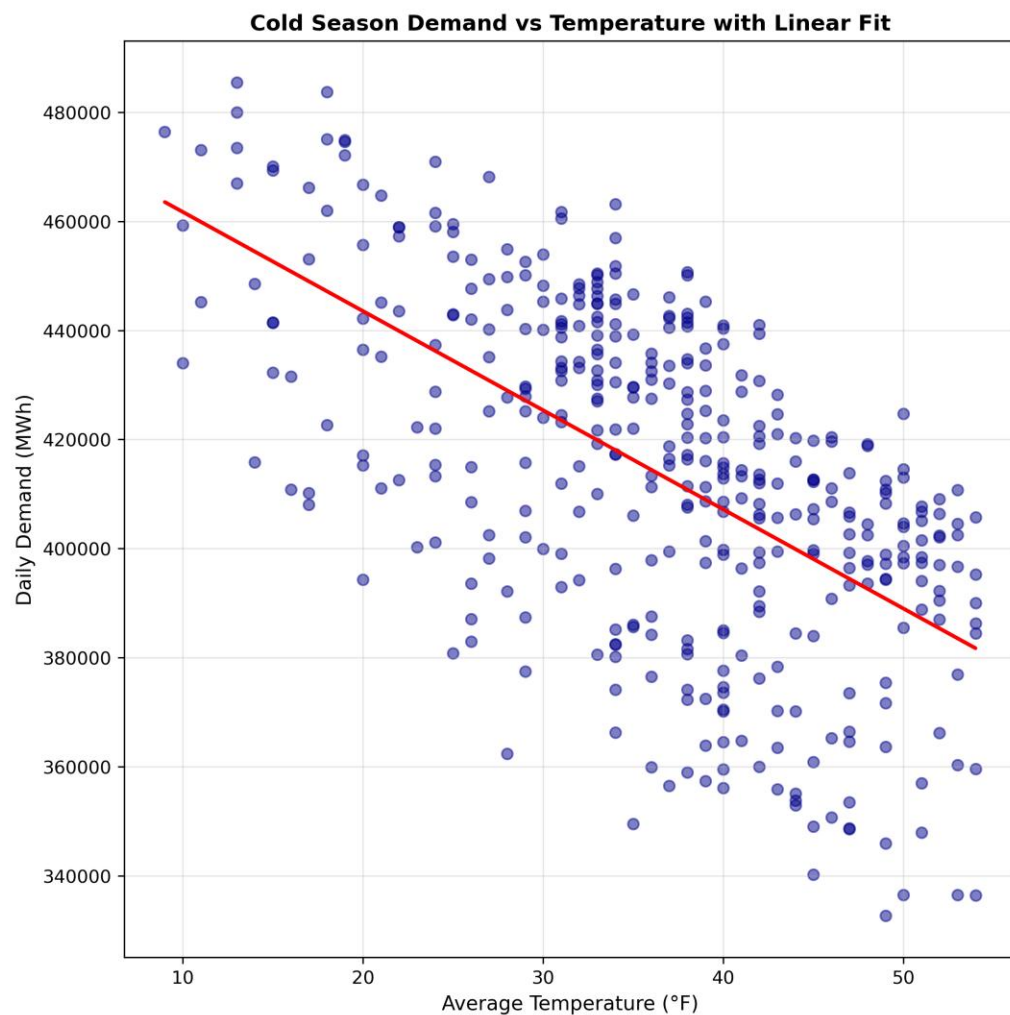
Linear Regression: Cold Season

Cold Season Test Results

RMSE: 24,754.41 MWh

MAE: 20,222.54 MWh

R^2 : 0.350



Linear Regression: Warm Season

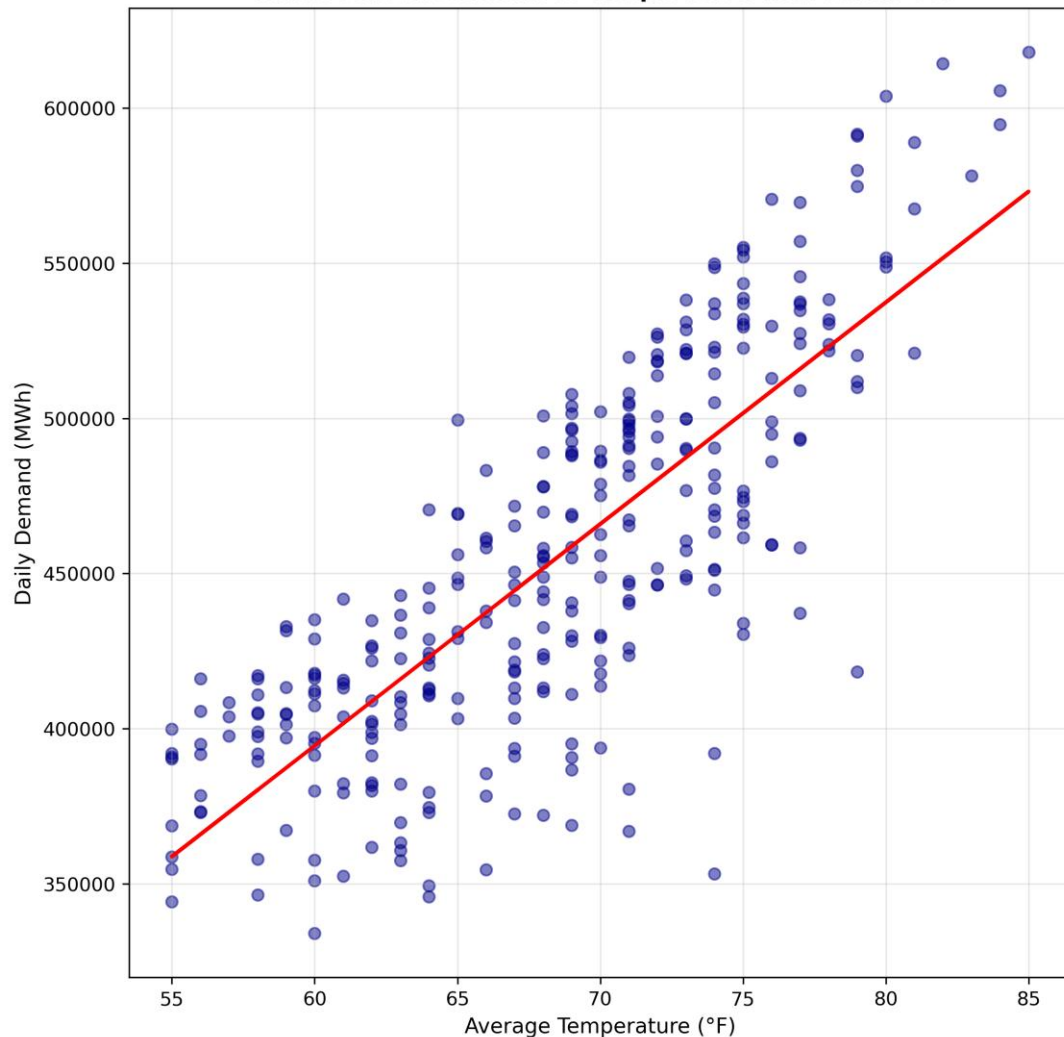
Warm Season Test Results

RMSE: 34,412.48 MWh

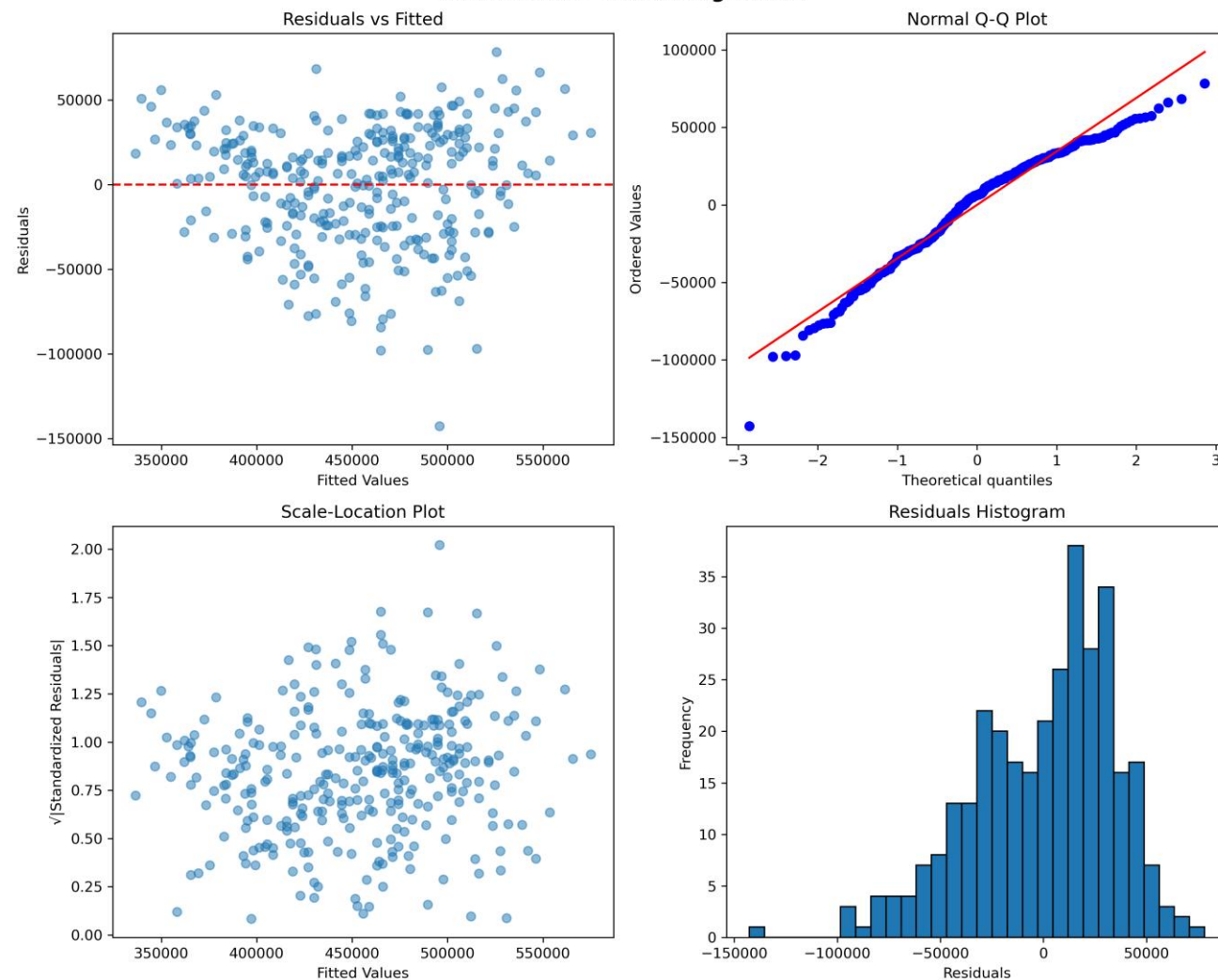
MAE: 27,806.15 MWh

R^2 : 0.684

Warm Season Demand vs Temperature with Linear Fit



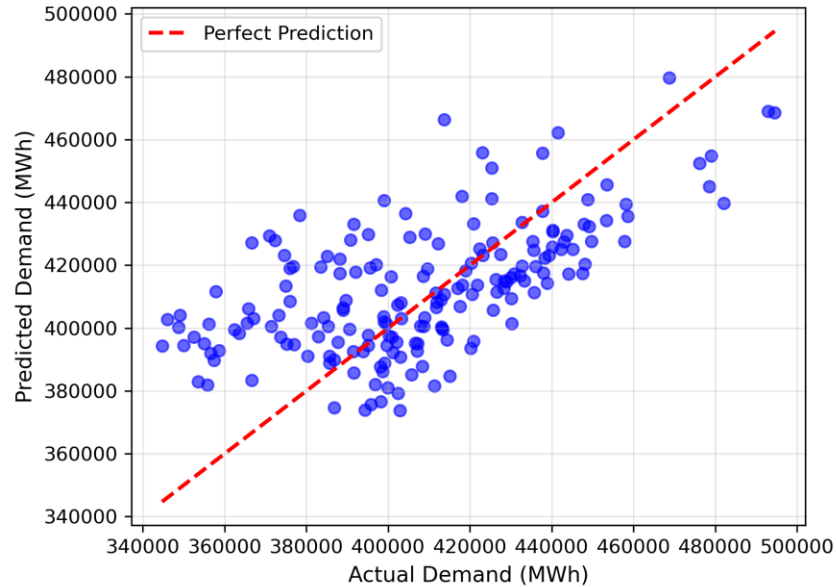
Warm Season Model Diagnostics



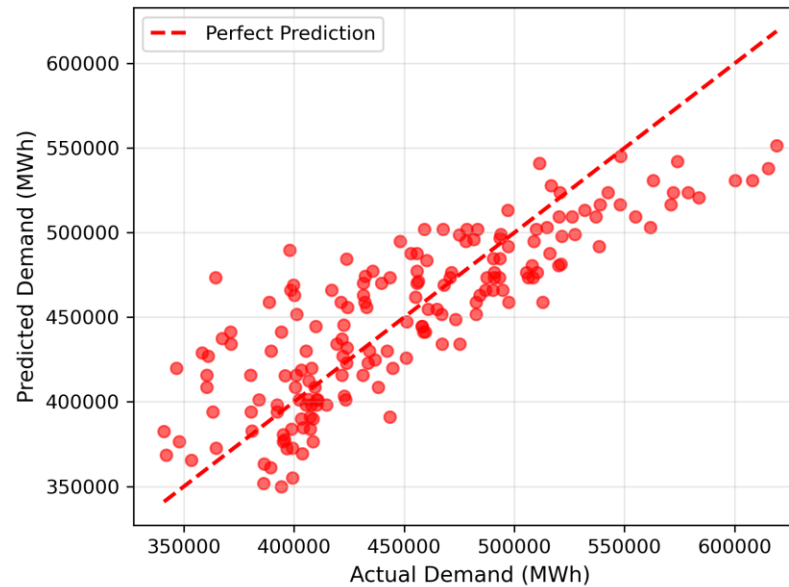
Linear Regression Predictions (2024 Test Data)

Linear Regression: Actual vs Predicted Demand (2024 Test Set)

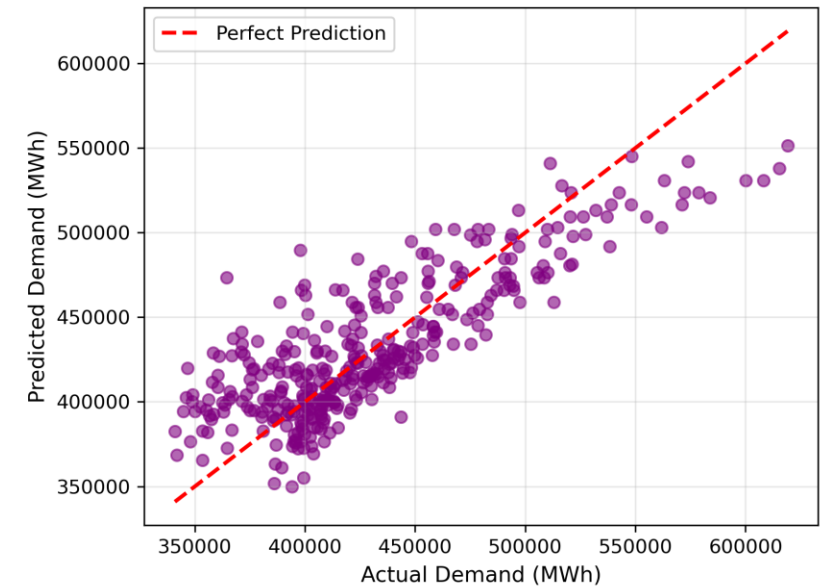
Cold Season ($R^2 = 0.350$)



Warm Season ($R^2 = 0.684$)



Overall 2024 ($R^2 = 0.679$)



Model Performance Summary

- **Warm Season Model (Avg Temp $\geq 55^{\circ}\text{F}$)**
 - **Predictors:** Average temperature, maximum humidity
 - **Training (2022-2023):** $R^2 = 0.665$, Adjusted $R^2 = 0.663$
 - **Testing (2024):** $R^2 = 0.68$
 - **Error:** RMSE = 34,412 MWh (~6% of typical demand)
- **Cold Season Model (Avg Temp $< 55^{\circ}\text{F}$)**
 - **Predictors:** Avg temp, precipitation, min humidity, windspeed, pressure
 - **Training (2022-2023):** $R^2 = 0.393$, Adjusted $R^2 = 0.385$
 - **Testing (2024):** $R^2 = 0.35$
 - **Error:** RMSE = 24,754 MWh (~5% of typical demand)

Current Model Limitations

**Predictive
Accuracy:**

Cold season model: $R^2 = 0.35$ (only 35% of demand explained by weather)

Warm season model underpredicts extreme demand (>525k MWh) where forecasting matters most

Multicollinearity:

High condition numbers
(cold: 12,200, warm: 1,540)

Natural correlation between weather variables difficult to eliminate

Data Constraints:

Statewide aggregated demand vs. single-location weather (Lansing Airport)

Daily aggregation loses hourly demand patterns

Limited to weather predictors only

Findings and Implications

Can Michigan's Grid Handle New Data Centers?

✓ Medium Data Centers (4 centers @ 50 MWh/day each):

- Additional load: 200 MWh/day
- **Assessment: Negligible impact** on grid (current range: 350k-600k MWh/day)

⚠ Hyperscale Data Centers (10 centers @ 1,400 MWh/day each):

- Additional load: 14,000 MWh/day (14 GWh)
- **Assessment: 2.3% added stress during peak demand** (days >600k MWh)

Recommendation: DTE & Consumers Energy should plan capacity increases for high temperature + high humidity days (>75°F + >80% humidity)

- Linear regression models predict typical weather patterns
- Further refinement needed for extreme demand prediction

Future Research Directions

Model Improvements:

- Add calendar variables (weekends, holidays, daylight hours)
- Compare polynomial regression or SARIMAX for better extreme value prediction
- Use seasonal rather than temperature thresholds

Enhanced Data:

- Hourly weather + demand data for more precise forecasting
- Location-specific substation demand (not statewide aggregates)

Vulnerability Analysis:

- Geospatial mapping: infrastructure age + transformer distance
- Social Vulnerability Index to identify at-risk populations during overload events

References

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- [2] Kelly House. 2025. Data centers eyed in at least 10 Michigan towns. How they might change state. *Bridge Michigan* (November 19, 2025). Retrieved December 2, 2025 from <https://bridgemi.com/michigan-environment-watch/data-centers-eyed-in-at-least-10-michigan-towns-how-they-might-change-state/>
- [3] Susan Vela. 2025. \$120 million data center proposed in downtown Lansing. What we know. *Lansing State Journal* (November 5, 2025). Retrieved December 2, 2025 from <https://www.lansingstatejournal.com/story/money/business/2025/11/05/lansing-data-center-deep-green-bwl/87051236007/>
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- [5] Ciaran Gilbert, Jethro Browell, and Bruce Stephen. 2023. Probabilistic load forecasting for the low voltage network: Forecast fusion and daily peaks. *Sustainable Energy, Grids and Networks* 34 (January 2023). <https://doi.org/10.1016/j.segan.2023.100998>
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- [9] Barry E. Flanagan, Edward W. Gregory, Elaine J. Hallisey, Janet L. Heitgerd, and Brian Lewis. 2011. A Social Vulnerability Index for Disaster Management. *Journal of Homeland Security and Emergency Management* 8, 1 (2011).
- [10] Karl Rabago and Robin Dutta. 2023. The Economic Impact of Michigan's Unreliable Power Grid. *Local Solar for All*. Retrieved from <https://house.mi.gov/Document/?DocumentId=39728&DocumentType=CommitteeTestimony>